

■ Series

`Series[f, {x, x0, n}]` generates a power series expansion for f about the point $x = x_0$ to order $(x - x_0)^n$.

`Series[f, {x, x0, nx}, {y, y0, ny}]` successively finds series expansions with respect to y , then x .

`Series` can construct standard Taylor series, as well as certain expansions involving negative powers, fractional powers and logarithms. ■ `Series` detects certain essential singularities. ■ `Series` can expand about the point $x = \infty$.

■ `Series[f, {x, 0, n}]` constructs Taylor series for any function f according to the formula $f(0) + f'(0)x + f''(0)x^2/2 + \dots + f^{(n)}(0)x^n/n!$. ■ `Series` effectively evaluates partial derivatives using `D`. It assumes that different variables are independent. ■ The result of `Series` is a `SeriesData` object, which you can manipulate with other functions. ■ See page 427. ■ See also: `InverseSeries`, `Limit`, `Normal`.